

****ICSE CLASS 10 PHYSICS — COMPLETE REVISION NOTES****

****CHAPTER 1: FORCE****

****YouTube Lecture:**** https://youtu.be/ONS_qjuKrq8?si=1bqyc_k2nqnRvu_-

****1. Turning Forces Concept and Moment of Force****

A force can produce two types of motion: translational motion (body moves as a whole) and rotational motion (body turns about a fixed point).

****Moment of Force (Torque):**** The turning effect produced by a force about a fixed point. It is the product of the force and the perpendicular distance from the axis of rotation to the line of action of the force.

Moment of Force = Force $\times$ Perpendicular Distance from the Axis

****SI Unit:**** Newton metre (N m). ****CGS Unit:**** dyne cm.

****Common Examples:**** Opening a door, turning a steering wheel, pedalling a bicycle, using a spanner to loosen a nut.

****Clockwise and Anticlockwise Moments:****

If the turning effect rotates the body clockwise, it is a clockwise moment. If the turning effect rotates the body anticlockwise, it is an anticlockwise moment.

****2. Principles of Equilibrium****

A body is said to be in equilibrium when no net force and no net moment act on it.

****Conditions for Equilibrium:****

The sum of all forces acting on the body in any direction is zero (translational equilibrium).

The sum of clockwise moments about a point equals the sum of anticlockwise moments about the same point (rotational equilibrium).

****Principle of Moments (Law of Lever):****

For a lever in equilibrium, Load $\times$ Load Arm = Effort $\times$ Effort Arm. This can be verified by suspending two spring balances with a metre rule and adjusting slotted weights.

****3. Centre of Gravity (Qualitative Understanding)****

The centre of gravity of a body is the point at which the entire weight of the body acts, irrespective of the orientation of the body.

For regular, uniform bodies (like a rod, circular disc, cube, sphere), the centre of gravity lies at their geometrical centre.

For an irregular lamina, the centre of gravity can be found by the plumb line method (intersection of lines of suspension).

****4. Uniform Circular Motion****

When a body moves along a circular path with constant speed, its velocity changes continuously because the direction changes. Therefore, there is an acceleration directed towards the centre of the circle.

****Centripetal Force:**** The force acting on a body moving in a circular path, directed towards the centre of the circle. It is necessary to keep the body in circular motion.

****Centrifugal Force:**** The equal and opposite reaction force experienced by the body, directed away from the centre. It is felt only in the rotating frame of reference.

****Difference:**** Centripetal force is real and acts towards the centre. Centrifugal force is a pseudo force that appears to act outwards in an accelerated frame.

****IMPORTANT QUESTIONS: FORCE****

****Q1.**** Define moment of force. State its SI and CGS units.

****Q2.**** State the two conditions for a body to be in equilibrium.

****Q3.**** State the principle of moments. Describe an experiment to verify it using a metre rule.

****Q4.**** A uniform metre rule is pivoted at its 50 cm mark. A weight of 20 gf is suspended at the 10 cm mark. Find the position where a weight of 40 gf should be suspended to balance the rule.

****Q5.**** What is meant by centre of gravity? Where does the centre of gravity of a uniform circular disc lie?

****Q6.**** Define uniform circular motion. Is the body moving with constant speed being accelerated? Give reason.

****Q7.**** Differentiate between centripetal force and centrifugal force.

****Q8.**** Why is it easier to open a door by applying force at its free end rather than near the hinge?

****Q9.**** A uniform metre rule of mass 100 g is balanced at the 40 cm mark when a weight of x gf is suspended at the 10 cm mark. Calculate x.

****Q10.**** Explain why a cyclist bends inwards while taking a curved turn.

****CHAPTER 2: WORK, ENERGY AND POWER****

****YouTube Lecture:**** <https://youtu.be/uhUTO6ilpZ4?si=NDKtDkkfXV6TIF-a>

****1. Work (Definition and Mathematical Expression)****

Work is said to be done when a force acting on a body produces a displacement in the direction of the force.

****Mathematical Expression:**** $W = FS \cos \theta$

Where F is the force, S is the displacement, and θ is the angle between the force and displacement vectors.

****Special Cases:****

When $\theta = 0^\circ$ (force and displacement in same direction): $W = FS$ (maximum positive work).

When $\theta = 90^\circ$ (force perpendicular to displacement): $W = 0$ (no work done). Example: carrying a bag while walking horizontally.

****SI Unit of Work:**** Joule (J). $1 \text{ J} = 1 \text{ N m}$. ****CGS Unit:**** erg. $1 \text{ J} = 10^7 \text{ erg}$.

Work done against gravity: $W = mgh$

****2. Energy (Definition and Types)****

Energy is the capacity to do work.

****A. Mechanical Energy****

****Kinetic Energy (K):**** The energy possessed by a body due to its motion.

$$K = \frac{1}{2} mv^2$$

****Potential Energy (U):**** The energy possessed by a body due to its position or configuration.

Gravitational Potential Energy: $U = mgh$

****B. Other Forms:**** Chemical energy (batteries, food), heat energy, electrical energy, nuclear energy, sound energy, light energy. Energy can change from one form to another (principle of energy conversion).

****3. Principle of Conservation of Energy****

Energy can neither be created nor destroyed. It can only be transformed from one form to another. The total energy of an isolated system always remains constant.

For a freely falling body, the total mechanical energy (Kinetic Energy + Potential Energy) remains constant throughout the fall.

****Application to a Simple Pendulum:****

At the extreme position: Maximum potential energy, zero kinetic energy.

At the mean (lowest) position: Maximum kinetic energy, minimum potential energy.

The total mechanical energy remains constant.

****4. Power****

Power is the rate of doing work.

****Expression:**** $P = W / t$

****SI Unit:**** Watt (W). $1 \text{ W} = 1 \text{ J s}^{-1}$. Larger units: kilowatt (kW), megawatt (MW), gigawatt (GW).

****Other Unit:**** Horsepower (hp). $1 \text{ hp} = 746 \text{ W}$.

****IMPORTANT QUESTIONS: WORK, ENERGY AND POWER****

****Q1.**** Define work. Write the expression for work done when a force F acts at an angle θ to the displacement S . When is work said to be zero?

****Q2.**** Derive the expressions for potential energy ($U = mgh$) and kinetic energy ($K = \frac{1}{2}mv^2$).

****Q3.**** State the principle of conservation of energy. Show that the total mechanical energy of a freely falling body is conserved.

****Q4.**** Explain the energy transformation in an oscillating simple pendulum.

****Q5.**** Define power. State the relationship between work, energy and power. Convert 1 hp into watts.

****Q6.**** A boy of mass 50 kg climbs a staircase of height 15 m in 30 seconds. Calculate the work done and power developed. ($g = 10 \text{ m/s}^2$)

****Q7.**** Differentiate between work and power. Give their SI units.

****Q8.**** Give one example each of the conversion of: (a) Chemical energy to electrical energy. (b) Electrical energy to mechanical energy.

****Q9.**** Name the type of energy possessed by a stretched bow and a moving arrow respectively.

****Q10.**** Calculate the kinetic energy of a car of mass 1000 kg moving with a velocity of 20 m/s.

****CHAPTER 3: MACHINES****

****YouTube Lecture:**** <https://youtu.be/4bghi4g6UhA?si=oE6xkgvsqrAqzcRc>

****1. Key Terms Related to Machines****

****Load (L):**** The resistance overcome by the machine.

****Effort (E):**** The force applied to the machine.

****Mechanical Advantage (MA):**** Ratio of load to effort. $(MA = L / E)$

****Velocity Ratio (VR):**** Ratio of velocity of effort to velocity of load. $(VR = V_E / V_L = d_E / d_L)$

****Efficiency (η):**** Ratio of output work to input work. $(\eta = W_o / W_i \times 100\%)$

****Relationship:**** $\eta = \text{MA} / \text{VR}$ (always less than 1 for a practical machine).

****2. Levers****

A lever is a rigid bar capable of turning about a fixed point called the fulcrum.

****Class I Lever (Fulcrum between Load and Effort):****

Examples: Seesaw, scissors, crowbar, pliers. Mechanical advantage can be $>$, $=$ or $<$ 1.

In the human body: nodding of the head (skull pivoting on the vertebral column).

****Class II Lever (Load between Fulcrum and Effort):****

Examples: Wheelbarrow, nutcracker, bottle opener. Mechanical advantage is always $>$ 1.

In the human body: standing on toes (ball of foot is fulcrum, body weight is load, calf muscle provides effort).

****Class III Lever (Effort between Fulcrum and Load):****

Examples: Broom, fishing rod, tweezers, sugar tongs. Mechanical advantage is always $<$ 1.

In the human body: lifting a load with the arm (elbow is fulcrum, biceps muscle provides effort, load in the palm).

****3. Pulley Systems****

****Single Fixed Pulley:**** Changes the direction of effort. $\text{MA} = 1$, $\text{VR} = 1$.

****Single Movable Pulley:**** $\text{MA} = 2$, $\text{VR} = 2$. Effort is applied upwards, load moves up.

****Block and Tackle System:**** A combination of fixed and movable pulleys. $\text{VR} =$ number of strands supporting the load. MA is slightly less than VR due to friction.

****IMPORTANT QUESTIONS: MACHINES****

****Q1.**** Define the terms: Mechanical Advantage, Velocity Ratio, and Efficiency. Derive the relationship between them.

****Q2.**** Explain why $\text{MA} < \text{VR}$ for all practical machines. How can you make an ideal machine?

****Q3.**** Name the three classes of levers. Give one example of each, including one from the human body. State the relative positions of fulcrum, load, and effort in each.

****Q4.**** Draw a neat labelled diagram of a single movable pulley. Why is its MA equal to 2?

****Q5.**** In a block and tackle system with 4 pulleys, a load of 400 N is raised by an effort of 120 N. Calculate (a) VR , (b) MA , (c) efficiency.

****Q6.**** Which class of lever always has $\text{MA} < 1$? Give an example.

****Q7.**** A man uses a crowbar to lift a stone. The length of the crowbar is 1.2 m and the fulcrum is placed 0.2 m from the stone. Calculate the effort needed to lift a load of 800 N, neglecting friction.

****Q8.**** Differentiate between a single fixed pulley and a single movable pulley.

****Q9.**** Explain why scissors for cutting cloth have long blades and short handles, while shears for cutting metals have short blades and long handles.

****Q10.**** A machine has $VR = 5$ and $MA = 4$. Calculate its efficiency. How much effort is required to overcome a load of 200 N?

****CHAPTER 4: REFRACTION OF LIGHT AT PLANE SURFACES****

****YouTube Lecture:**** <https://youtu.be/5AVICIOOJDs?si=TzcOySn-CcXk7CxH>

****1. Introduction to Refraction****

Refraction is the change in the direction of light when it passes from one transparent medium to another optically different medium.

****Cause:**** Change in the speed of light in different media.

****Effect on Wave Properties:**** Frequency remains constant. The speed (v) and wavelength (λ) change. $v = f\lambda$.

****2. Laws of Refraction****

****First Law:**** The incident ray, the refracted ray and the normal at the point of incidence all lie in the same plane.

****Second Law (Snell's Law):**** For a given pair of media, the ratio of the sine of the angle of incidence (i) to the sine of the angle of refraction (r) is a constant.

$$\left(\frac{\sin i}{\sin r} = \right) \text{constant} = \text{Refractive Index } (\mu)$$

****3. Refractive Index****

****Absolute Refractive Index:**** The ratio of the speed of light in vacuum (c) to the speed of light in the medium (v).

$$\left(\mu = \frac{c}{v} \right)$$

****Relative Refractive Index:**** The ratio of the speed of light in medium 1 to the speed of light in medium 2.

Values: Water (1.33), Glass (1.5), Diamond (2.42).

****Condition for No Deviation:**** A ray entering a new medium does not bend if it strikes the surface at normal incidence ($i = 0$).

****4. Refraction through a Glass Block (Rectangular Slab)****

The ray bends towards the normal as it enters the slab (rarer to denser).

The ray bends away from the normal as it exits the slab (denser to rarer).

The emergent ray is parallel to the incident ray but undergoes a **lateral displacement**.

Lateral displacement increases with increasing thickness, angle of incidence, and refractive index.

****5. Real and Apparent Depth****

When an object is placed inside a denser medium (like water), it appears to be at a shallower depth when viewed from a rarer medium (like air).

****Real Depth:**** The actual depth of the object from the surface.

****Apparent Depth:**** The depth at which the object appears.

$$\mu = \frac{\text{Real Depth}}{\text{Apparent Depth}}$$

This explains why a coin at the bottom of a beaker filled with water appears raised, why ponds appear shallower, and why a stick partly immersed in water appears bent.

****6. Total Internal Reflection****

When light travels from a denser medium to a rarer medium, it bends away from the normal. As the angle of incidence increases, at a certain angle called the critical angle (C), the angle of refraction becomes 90° .

For any angle of incidence greater than the critical angle, the light is completely reflected back into the denser medium. This phenomenon is called Total Internal Reflection.

****Essential Conditions:****

Light must travel from a denser medium to a rarer medium.

The angle of incidence in the denser medium must be greater than the critical angle for that pair of media.

****Relationship:**** $\mu = \frac{1}{\sin C}$

****7. Applications of Total Internal Reflection****

****Triangular Glass Prisms (Right Angled, 45° , 45° , 90°):****

Used to deviate a ray through 90° .

Used to deviate a ray through 180° (erect image in inverted position).

Used to invert an image without deviation (binoculars).

Comparison with Plane Mirror: Total internal reflection from a prism gives almost 100% reflection without loss of light, unlike a silvered plane mirror where some light is absorbed.

****IMPORTANT QUESTIONS: REFRACTION OF LIGHT AT PLANE SURFACES****

****Q1.**** Define refraction of light. State the two laws of refraction.

****Q2.**** Define refractive index. Write its formula relating to the speed of light in vacuum.

****Q3.**** A ray of light enters a glass slab at an angle of 45° to the normal. Explain its path with a labelled diagram. State the nature of the emergent ray.

****Q4.**** What is lateral displacement? State two factors on which it depends.

****Q5.**** Explain why a coin placed at the bottom of a beaker filled with water appears raised. Define real and apparent depth.

****Q6.**** What is total internal reflection? State the two essential conditions for it to occur. Define critical angle.

****Q7.**** With the help of a ray diagram, explain how a right angled isosceles prism (45° , 90° , 45°) can deviate a ray through 90° .

****Q8.**** Why is a prism preferred over a plane mirror for reflecting light in periscopes?

****Q9.**** Calculate the critical angle for a medium whose refractive index is $\sqrt{2}$.

****Q10.**** State the factor on which the refractive index of a medium depends (other than the nature of the medium itself).

****CHAPTER 5: REFRACTION THROUGH A LENS****

****YouTube Lecture:**** https://youtu.be/-A0yLShb85k?si=F_NF3Efyw94yCKqS

****1. Introduction to Lenses****

A lens is a transparent refracting medium bounded by two spherical surfaces.

****Convex (Converging) Lens:**** Thicker in the centre, thinner at the edges. It converges parallel rays to a point.

****Concave (Diverging) Lens:**** Thinner in the centre, thicker at the edges. It diverges parallel rays.

****Key Terms:****

****Pole (O):**** The geometrical centre of the spherical surface of the lens.

****Principal Axis:**** The line joining the centres of curvature of the two surfaces.

****Optical Centre (C):**** The central point of the lens where a ray passes undeviated.

****Focus (F):**** The point on the principal axis where parallel rays converge (convex) or appear to diverge from (concave).

****Focal Length (f):**** Distance between the optical centre and the focus.

****2. Image Formation by Convex Lens (Ray Diagrams)****

Position of Object	Image Position	Size	Nature
At infinity	At F	Highly diminished (point)	Real and inverted
Beyond 2F	Between F and 2F	Diminished	Real and inverted
At 2F	At 2F	Same size	Real and inverted
Between F and 2F	Beyond 2F	Magnified	Real and inverted
At F	At infinity	Highly magnified	Real and inverted
Between F and O	On same side	Magnified	Virtual and erect

****Uses of Convex Lens:**** Magnifying glass (simple microscope), camera, human eye lens, projector, microscope.

****3. Image Formation by Concave Lens****

For a concave lens, regardless of the object position, the image is always virtual, erect, diminished, and located on the same side as the object between the optical centre and the focus.

****4. Sign Convention (New Cartesian / Lens Formula Convention)****

Optical centre is the origin. Distances measured in the direction of incident light are positive. Distances measured opposite to incident light are negative. Heights above the principal axis are positive; below are negative.

****5. Lens Formula and Magnification****

****Lens Formula:**** $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$

****Magnification:**** $m = \frac{v}{u} = \frac{\text{Height of image}}{\text{Height of object}}$

****6. Power of a Lens****

Power is the ability of a lens to converge or diverge a beam of light. It is the reciprocal of focal length in metres.

$P = \frac{1}{f}$ (in metres)

****SI Unit:**** Dioptre (D). Convex lens has positive power; concave lens has negative power.

****IMPORTANT QUESTIONS: REFRACTION THROUGH A LENS****

****Q1.**** Differentiate between a convex lens and a concave lens. Draw their symbols.

****Q2.**** With the help of ray diagrams, show the formation of the image by a convex lens when the object is placed (a) beyond 2F, (b) at 2F, and (c) between F and O.

****Q3.**** State the nature and position of the image formed by a convex lens when the object is at F.

****Q4.**** Where must an object be placed before a convex lens to obtain a real image of the same size? Write your answer in terms of the lens formula.

****Q5.**** An object 5 cm high is placed at a distance of 20 cm from a convex lens of focal length 15 cm. Find the position, nature and size of the image.

****Q6.**** State one use each of a convex lens and a concave lens.

****Q7.**** A virtual, erect and diminished image is formed by a lens. Name the type of lens. Draw its ray diagram.

****Q8.**** Define power of a lens. State its SI unit. A convex lens has a focal length of 25 cm. Calculate its power.

****Q9.**** What is the sign convention for convex and concave lenses?

****Q10.**** How does a lens act as a combination of prisms? Explain.

****CHAPTER 6: SPECTRUM****

****YouTube Lecture:**** https://youtu.be/Vk1BaG_lbj0?si=Dj6t5smqA4R9AZMo

****1. Dispersion of Light Through a Prism****

Dispersion is the splitting of white light into its constituent colours when passed through a triangular prism. The band of seven colours obtained is called a spectrum. Order of colours: Violet, Indigo, Blue, Green, Yellow, Orange, Red (VIBGYOR).

****Cause:**** The refractive index of glass is different for different colours (wavelengths). Violet light has the shortest wavelength and suffers maximum deviation. Red light has the longest wavelength and suffers minimum deviation.

****2. The Electromagnetic Spectrum****

The entire range of electromagnetic radiations arranged in order of increasing wavelength.

****Complete Arrangement (Increasing Wavelength):****

Gamma rays → X rays → Ultraviolet (UV) rays → Visible Light → Infrared (IR) rays → Microwaves → Radio waves

****Properties Common to All Electromagnetic Waves:****

All travel at the speed of light in vacuum (3×10^8 m/s).

All are transverse waves.

They do not require a medium for propagation.

****Properties and Uses of Specific Radiations:****

****Infrared (IR) Rays:**** Thermal radiation (heat). Uses: Remote controls, night vision cameras, therapeutic heating, taking photographs in fog/haze.

****Ultraviolet (UV) Rays:**** More energetic. Uses: Sterilising medical equipment and water, detecting forged documents and fingerprints, source of Vitamin D (but excess causes skin cancer).

****3. Scattering of Light****

When sunlight passes through the atmosphere, tiny particles scatter the light. The amount of scattering depends on the wavelength of light.

Shorter wavelengths (blue/violet) are scattered much more than longer wavelengths (red/orange).

****Blue Colour of the Sky:**** Blue light is scattered the most by atmospheric particles, so the sky appears blue.

****Red Appearance of Sun at Sunrise/Sunset:**** At the horizon, sunlight passes through a greater thickness of air. Most of the blue light is scattered away, and only the red light (least scattered) reaches our eyes.

****IMPORTANT QUESTIONS: SPECTRUM****

****Q1.**** What is dispersion of light? Name the phenomenon that causes it. Draw a diagram of the dispersion of white light by a prism.

****Q2.**** Explain why violet light deviates the most and red light deviates the least when passing through a prism.

****Q3.**** What is the electromagnetic spectrum? Arrange the following in increasing order of wavelength: Infrared, X-rays, Gamma rays, Radio waves, Ultraviolet, Microwaves.

****Q4.**** State any two properties common to all electromagnetic radiations.

****Q5.**** State two uses each of: (a) Ultraviolet rays, (b) Infrared rays.

****Q6.**** Explain the scattering of light. Why does the sky appear blue?

****Q7.**** Explain why the sun appears red at sunrise and sunset.

****Q8.**** Name the electromagnetic radiation used: (a) in remote controls, (b) to kill germs in water purifiers.

****Q9.**** What would be the colour of the sky if Earth had no atmosphere? Give reason.

****Q10.**** An astronaut in space sees the sky as black. Give reason.

****CHAPTER 7: SOUND****

****YouTube Lecture:**** <https://youtu.be/tme78Q0ITkY?si=9BBftoMLvAIDwWAX>

****1. Nature and Propagation of Sound****

Sound is a form of energy that produces a sensation of hearing. It travels in the form of longitudinal waves.

Sound requires a material medium (solid, liquid, or gas) to travel. It cannot travel through a vacuum.

Speed of sound: Solids > Liquids > Gases.

****Comparison with Light:**** Light travels much faster (3×10^8 m/s) than sound (340 m/s in air). This explains why thunder is heard after the flash of lightning is seen.

****2. Key Terms Related to Sound Waves****

****Frequency (f):**** Number of oscillations per second. SI unit: Hertz (Hz).

****Wavelength (λ):**** Distance covered by one complete wave. Unit: metre (m).

****Velocity (v):**** Speed of the wave. Relation: $v = f\lambda$

****3. Classification of Sound Based on Frequency****

****Infrasonic Sound:**** Frequency below 20 Hz. Inaudible to humans. Produced by earthquakes, elephants.

****Sonic (Audible) Sound:**** Frequency between 20 Hz and 20,000 Hz. Normal audible range for humans.

****Ultrasonic Sound:**** Frequency above 20,000 Hz. Inaudible to humans. Produced by bats, dogs. Uses: SONAR, medical imaging (ultrasonography), cleaning delicate parts, detecting flaws in metals.

****Supersonic Speed:**** The speed of an object greater than the speed of sound (not a frequency range).

****4. Reflection of Sound (Echo)****

Sound waves bounce back when they hit a hard surface. The repetition of sound due to reflection is called an echo.

****Conditions for a Distinct Echo:**** The minimum distance between the source and the reflecting surface should be 17 m (for persistence of hearing of 0.1 s).

****Reverberation:**** The persistence of sound due to multiple reflections in a closed room.

****Uses of Echoes:**** Bats navigate and detect prey using echoes of ultrasound. SONAR (Sound Navigation and Ranging) uses ultrasonic echoes to measure the depth of the sea and locate underwater objects. Used in medical fields (ultrasonography) to check internal organs and foetal development.

****5. Characteristics of Sound****

****Loudness:**** Related to the amplitude of the vibration. Larger amplitude means louder sound. The unit of sound level is the decibel (dB). Intensity is the amount of sound energy per unit area. Loudness is the subjective sensation perceived by the ear.

****Pitch:**** Related to the frequency of the vibration. Higher frequency gives a shriller sound (higher pitch). Lower frequency gives a flat, base sound (lower pitch).

****Quality (Timbre):**** The characteristic that allows us to distinguish between notes of the same loudness and pitch produced by different instruments. It depends on the waveform (shape of the sound wave).

****6. Vibrations and Resonance****

****Natural (Free) Vibrations:**** Vibrations of a body in the absence of any external periodic force. Frequency = natural frequency.

****Damped Vibrations:**** The amplitude of vibration decreases with time due to friction or resistance.

****Forced Vibrations:**** Vibrations of a body under the influence of an external periodic force. The body vibrates with the frequency of the external force, not its own.

****Resonance:**** A special case of forced vibration where the frequency of the external force matches the natural frequency of the body. The body vibrates with a large amplitude. Examples: Tuning fork, breaking a glass with a high pitched note, soldiers breaking step while crossing a bridge.

****IMPORTANT QUESTIONS: SOUND****

****Q1.**** Why can sound not travel through a vacuum? What type of wave is a sound wave in air?

****Q2.**** An observer hears thunder 3 seconds after seeing a lightning flash. How far is the lightning from the observer? (Speed of sound in air = 340 m/s)

****Q3.**** State the relationship between wave velocity, frequency and wavelength for sound waves. Calculate the wavelength of a sound wave of frequency 500 Hz and speed 340 m/s.

****Q4.**** Define the following: (a) Infrasonic sound, (b) Ultrasonic sound. Give one use of ultrasonic sound.

****Q5.**** What is SONAR? Name the type of waves used in SONAR and state its principle.

****Q6.**** What is an echo? State the minimum distance between the source and the reflector required to hear a distinct echo.

****Q7.**** Differentiate between loudness and pitch of a sound on the basis of the physical quantity involved.

****Q8.**** Distinguish between natural vibrations, forced vibrations, and resonance. Give one example of resonance.

****Q9.**** What is the difference between ultrasonic and supersonic?

****Q10.**** Why are the walls of a cinema hall covered with sound absorbing materials?

****CHAPTER 8: CALORIMETRY****

****YouTube Lecture:**** https://youtu.be/ZHG30f6aH5U?si=t_ff5hoSNDr2p1oT

****1. Heat and Temperature****

Heat is a form of energy that flows from a body at a higher temperature to a body at a lower temperature. SI unit: Joule (J).

Temperature is the degree of hotness or coldness of a body. It determines the direction of heat flow. Units: Celsius ($^{\circ}\text{C}$) and Kelvin (K).

****2. Thermal (Heat) Capacity****

The amount of heat energy required to raise the temperature of a given body by 1°C (or 1 K).

$$C' = Q / \Delta T$$

SI unit: Joule per Kelvin (J/K) or Joule per $^{\circ}\text{C}$.

****3. Specific Heat Capacity****

The amount of heat energy required to raise the temperature of unit mass (1 kg) of a substance by 1°C (or 1 K).

$$c = Q / (m \times \Delta T)$$

SI unit: Joule per kilogram per Kelvin ($\text{J kg}^{-1} \text{K}^{-1}$). ****Relation with Heat Capacity:**** $C' = mc$

****Values:**** Water ($4200 \text{ J kg}^{-1} \text{K}^{-1}$), Ice ($2100 \text{ J kg}^{-1} \text{K}^{-1}$), Copper ($400 \text{ J kg}^{-1} \text{K}^{-1}$).

****Importance of High Specific Heat of Water:**** Water can store a large amount of heat with a small rise in temperature. This is why water is used as a coolant in car radiators, and why coastal areas have a moderate climate (sea water heats up and cools down slowly).

****4. Principle of Calorimetry (Method of Mixtures)****

When two bodies at different temperatures are brought into thermal contact, heat flows from the hotter body to the colder body until they reach the same temperature.

****Mathematical Statement:**** Heat Lost by Hot Body = Heat Gained by Cold Body (assuming no heat loss to the surroundings).

****5. Latent Heat****

The heat energy required to change the state (phase) of a substance without changing its temperature.

****Specific Latent Heat of Fusion (L):**** The amount of heat required to convert 1 kg of a solid into liquid at its melting point without any change in temperature. Unit: J/kg. Example for ice: 336,000 J/kg.

****Heating Curve for Water:****

Ice at $T < 0^{\circ}\text{C}$ absorbs heat, temperature rises. At 0°C , ice melts (temperature constant, latent heat absorbed). Water at 0°C absorbs heat, temperature rises. At 100°C , water boils (temperature constant, latent heat absorbed). Steam at $T > 100^{\circ}\text{C}$ absorbs heat, temperature rises.

****IMPORTANT QUESTIONS: CALORIMETRY****

****Q1.**** Define the term specific heat capacity. Write its SI unit. Differentiate between thermal capacity and specific heat capacity.

****Q2.**** Explain, with an example, why water is used as a coolant in car radiators.

****Q3.**** Why does ice at 0°C produce a greater cooling effect than water at 0°C ?

****Q4.**** 200 g of hot water at 80°C is mixed with 300 g of cold water at 20°C . Calculate the final temperature of the mixture. (Specific heat capacity of water = $4.2 \text{ J/g}^{\circ}\text{C}$)

****Q5.**** State the principle of calorimetry.

****Q6.**** Define specific latent heat of fusion. State its SI unit.

****Q7.**** A container holding 500 g of water at 25°C is heated. The water reaches 75°C by absorbing 105 kJ of heat. Calculate the specific heat capacity of water from this data.

****Q8.**** Explain the terms melting, boiling and latent heat with reference to the heating curve of water.

****Q9.**** 50 g of ice at 0°C is mixed with 200 g of water at 40°C . Find the final temperature. (Specific latent heat of fusion of ice = 336 J/g , specific heat capacity of water = $4.2 \text{ J/g}^{\circ}\text{C}$)

****Q10.**** What happens to the temperature of a substance during change of state? Give reason.

****CHAPTER 9: CURRENT ELECTRICITY****

****YouTube Lecture:**** <https://youtu.be/mvZKUWQBuOc?si=Q7c51423IAFSB-kQ>

****1. Basic Concepts****

****Electric Current (I):**** The rate of flow of electric charge (Q) through a cross section of a conductor.

$$I = Q / t$$

SI unit: Ampere (A). 1 A = 1 C/s. In metals, current is due to the flow of electrons (conventional current direction is opposite to electron flow).

****Potential Difference (V):**** The work done (W) in moving a unit charge (Q) between two points in a circuit.

$$V = W / Q$$

SI unit: Volt (V). 1 V = 1 J/C.

****Electromotive Force (emf):**** The total energy supplied by a cell in moving a unit charge around a complete circuit. Also measured in Volt.

****2. Ohm's Law****

For a given conductor, the current (I) flowing through it is directly proportional to the potential difference (V) applied across its ends, provided its temperature and other physical conditions remain constant.

$$V = IR$$

****Resistance (R):**** The opposition offered by a conductor to the flow of current. SI unit: Ohm (Ω). 1 Ω = 1 V/A.

****Verification:**** Plot a graph of V vs I. It should be a straight line passing through origin. Slope = Resistance.

****Ohmic Resistors:**** Follow Ohm's law (e.g., metals). ****Non Ohmic Resistors:**** Do not follow Ohm's law (e.g., diodes, LED, thyristor).

****3. Factors Affecting Resistance****

Resistance (R) is directly proportional to the length (l) of the conductor. $R \propto l$

Resistance is inversely proportional to the cross sectional area (A). $R \propto 1/A$

Resistance depends on the nature of the material (resistivity ρ). $R = \rho l / A$

Resistance of a metal increases with temperature. Resistance of a semiconductor decreases with temperature.

****4. Resistances in Series and Parallel****

****Series Combination:****

Same current flows through each resistor.

Total potential difference: $V = V_1 + V_2 + V_3$

Equivalent resistance: $R_{eq} = R_1 + R_2 + R_3$

****Parallel Combination:****

Potential difference across each resistor is the same.

Total current: $(I = I_1 + I_2 + I_3)$

Equivalent resistance: $(1/R_{eq} = 1/R_1 + 1/R_2 + 1/R_3)$

Equivalent resistance is always less than the smallest individual resistance.

****IMPORTANT QUESTIONS: CURRENT ELECTRICITY****

****Q1.**** State Ohm's law. Describe an experiment with a circuit diagram to verify it. Draw the V-I graph for an ohmic resistor.

****Q2.**** Define resistance. State its SI unit. Name two factors on which the resistance of a conductor depends.

****Q3.**** Derive an expression for the equivalent resistance of three resistors connected in (a) series, and (b) parallel.

****Q4.**** Calculate the equivalent resistance when three resistors of $2\ \Omega$, $3\ \Omega$ and $6\ \Omega$ are connected in parallel.

****Q5.**** A wire of resistance $10\ \Omega$ is stretched to twice its original length. What is its new resistance?

****Q6.**** Differentiate between emf of a cell and terminal potential difference.

****Q7.**** Define potential difference between two points in a circuit. Write its relation with work done and charge.

****Q8.**** A bulb is rated at $5\ \text{V}$, $500\ \text{mA}$. Calculate its power and resistance.

****Q9.**** What is the difference between an ohmic and a non ohmic resistor? Give an example of each.

****Q10.**** In a circuit, a cell of emf $2\ \text{V}$ and internal resistance $0.1\ \Omega$ is connected across an external resistance of $4.9\ \Omega$. Calculate the current drawn and the terminal voltage of the cell.

****CHAPTER 10: ELECTRICAL POWER AND HOUSEHOLD CIRCUITS****

****YouTube Lecture:**** <https://youtu.be/C4ZWFIXvwnk?si=JksS0PG-Pqe9Iama>

****1. Electrical Energy and Power****

****Electrical Energy (W):**** The work done by a source in driving current through a conductor.
($W = QV = VIt = I^2Rt = (V^2 / R) t$)

****Electrical Power (P):**** The rate of doing electrical work.

($P = W / t = VI = I^2R = V^2 / R$)

****SI Unit of Power:**** Watt (W). ****Larger Units:**** Kilowatt (kW), Megawatt (MW), Gigawatt (GW).

****Commercial Unit of Electrical Energy:**** Kilowatt hour (kW h). $1 \text{ kW h} = 3.6 \times 10^6 \text{ J}$. This is also called one "unit" or Board of Trade Unit (BOTU).

****Power Rating of Appliances:**** Higher wattage means more energy consumed per second.

****2. Household Wiring (Ring System)****

The main supply cable enters the house and goes to the kWh meter and then to the main fuse. Then it is connected to a distribution board. A ring main circuit is used for sockets and a separate radial circuit is used for lights.

****Three Wires in Domestic Supply:****

****Live Wire (L):**** Brown or Red, at 220 V AC.

****Neutral Wire (N):**** Light Blue or Black, at 0 V.

****Earth Wire (E):**** Green or Yellow/Green, connected to a copper plate buried deep in the ground. Earth wire provides a low resistance path for current in case of a fault.

****Main Fuse and MCB:**** All appliances are wired in parallel so each gets 220V and can be operated independently.

****Staircase Wiring (Two Way Switch):**** A bulb is controlled by two switches placed at two different positions (e.g., top and bottom of stairs).

****3. Safety Precautions****

****Fuse:**** A safety device made of an alloy (tin + lead) with low melting point. It melts and breaks the circuit if current exceeds a safe value. It is always placed in the live wire.

****Earthing:**** Protects the user from electric shock. If a live wire touches the metal body of an appliance, current flows through the earth wire instead of the user.

****Three Pin Plug and Socket:**** Pins for live, neutral and earth. Earth pin is longer and thicker to connect first and disconnect last.

****Colour Coding:**** Red (Live), Black (Neutral), Green (Earth). International: Brown (Live), Light Blue (Neutral), Green/Yellow (Earth).

Switches and fuses must always be connected in the live wire.

****IMPORTANT QUESTIONS: ELECTRICAL POWER AND HOUSEHOLD CIRCUITS****

****Q1.**** Define electrical power. Write its SI unit and its relation with voltage and current.

****Q2.**** An electric heater of resistance $50\ \Omega$ is connected across 220 V. Calculate (a) current drawn, (b) power of the heater, and (c) energy consumed in 5 hours in kWh.

****Q3.**** What is the commercial unit of electrical energy? Derive its relation with the joule.

****Q4.**** Explain the ring system of household wiring. Why are appliances connected in parallel?

****Q5.**** Name the three wires used in domestic wiring. State their colour codes and functions.

****Q6.**** What is the function of an earth wire? State the principle of earthing and why it is a safety measure.

****Q7.**** Why is a fuse used in an electric circuit? Why is it always connected to the live wire?

****Q8.**** Draw a labelled circuit diagram for staircase wiring using two way switches.

****Q9.**** Differentiate between a fuse wire and a switch.

****Q10.**** A house has the following loads: 4 bulbs of 60 W each for 5 hours, 2 fans of 100 W each for 8 hours, and 1 AC of 2 kW for 4 hours. Calculate the total energy consumed in a month (30 days) in kWh.

****CHAPTER 11: ELECTROMAGNETISM****

****YouTube Lecture:**** <https://youtu.be/277lWoUr36E?si=jNRMSRJ-d15woj3i>

****1. Magnetic Effect of Electric Current (Oersted's Discovery)****

When an electric current passes through a straight conductor, a magnetic field is produced around it. The direction of the magnetic field is given by the Right Hand Thumb Rule: If the thumb points in the direction of current, the curled fingers give the direction of the magnetic field lines.

****2. Magnetic Field Due to a Current Carrying Loop and Solenoid****

A coil with many turns of insulated wire is called a solenoid. When current flows, it behaves like a bar magnet. The magnetic field inside a solenoid is uniform and parallel. An electromagnet is a solenoid with a soft iron core. It is a temporary magnet (magnetism lasts as long as current flows).

****Comparison of Electromagnet and Permanent Magnet:****

Electromagnet: Magnetism is temporary, strength can be varied, polarity can be reversed.

Permanent Magnet: Magnetism is permanent, strength is fixed, polarity cannot be reversed.

****3. The DC Electric Motor****

****Principle:**** A current carrying conductor placed in a magnetic field experiences a force. The direction of force is given by Fleming's Left Hand Rule.

****Fleming's Left Hand Rule:**** Stretch the forefinger, middle finger and thumb of the left hand mutually perpendicular. Forefinger = Direction of magnetic Field. Middle Finger = Direction of Current. Thumb = Direction of Motion (Force) on the conductor.

****Construction:**** Permanent magnet (field), Armature coil, Split ring commutator, Brushes.

****Function of Split Ring Commutator:**** Reverses the direction of current in the coil every half rotation, ensuring that the coil continues to rotate in the same direction.

****Energy Conversion:**** Electrical energy to mechanical (kinetic) energy.

****4. Electromagnetic Induction****

The phenomenon by which an emf (and hence current) is induced in a closed coil when there is a change in the magnetic flux linked with the coil. Discovered by Michael Faraday.

****Fleming's Right Hand Rule (Dynamo Rule):**** Stretch the forefinger, middle finger and thumb of the right hand mutually perpendicular. Forefinger = Direction of magnetic Field. Thumb = Direction of Motion of conductor. Middle Finger = Direction of Induced Current.

****5. The AC Generator****

****Principle:**** Electromagnetic induction.

****Construction:**** Similar to motor, but uses Slip Rings instead of split ring commutator.

****Function of Slip Rings:**** Provide continuous contact, allowing the current to flow in the external circuit in alternating directions.

****Energy Conversion:**** Mechanical energy to electrical energy.

****Frequency of AC in India:**** 50 Hz.

****Advantage of AC over DC:**** AC can be easily stepped up or stepped down using a transformer, which minimises energy loss during long distance transmission.

****6. Transformer****

A device used to change the voltage of an alternating current. Works on the principle of mutual induction.

****Step Up Transformer:**** Secondary voltage > Primary voltage. $(N_s > N_p)$

****Step Down Transformer:**** Secondary voltage < Primary voltage. $(N_s < N_p)$

$(V_s / V_p = N_s / N_p)$

Used in power stations to step up voltage for transmission and step down for domestic use.

****IMPORTANT QUESTIONS: ELECTROMAGNETISM****

****Q1.**** State the Right Hand Thumb Rule. Draw the magnetic field lines around a straight current carrying conductor.

****Q2.**** What is an electromagnet? Give two differences between an electromagnet and a permanent magnet.

****Q3.**** State Fleming's Left Hand Rule. Name a device that uses this rule.

****Q4.**** With a labelled diagram, describe the construction and working of a DC electric motor. State the function of the split ring commutator.

****Q5.**** What is electromagnetic induction? State Faraday's law.

****Q6.**** State Fleming's Right Hand Rule. Name a device working on the principle of electromagnetic induction.

****Q7.**** With a labelled diagram, explain the construction and working of an AC generator. Why are slip rings used instead of a split ring commutator?

****Q8.**** State two advantages of AC over DC for long distance power transmission.

****Q9.**** Explain the working principle of a transformer. Draw a labelled diagram of a step up transformer.

****Q10.**** A transformer has 50 turns in its primary coil and 500 turns in its secondary coil. If the primary voltage is 220 V, what is the secondary voltage? Name the type of transformer.

****CHAPTER 12: RADIOACTIVITY****

****YouTube Lecture:**** <https://youtu.be/V6hYbriPhqk?si=E3L6Ql82Mxeu2ce8>

****1. Structure of the Nucleus****

An atom consists of a tiny, dense, positively charged nucleus containing protons and neutrons. Electrons revolve around the nucleus.

****Atomic Number (Z):**** Number of protons in the nucleus. Determines the element.

****Mass Number (A):**** Total number of protons and neutrons (nucleons) in the nucleus.

****Nuclear Structure Equation:**** $A = Z + N$ (where N is the number of neutrons).

****2. Radioactivity****

Radioactivity is the phenomenon of spontaneous disintegration of the nuclei of heavy elements with the emission of alpha (α), beta (β) and gamma (γ) radiations.

****Nature and Properties of Radiations:****

Property	Alpha (α) Particle	Beta (β) Particle	Gamma (γ) Ray
Nature	Helium nucleus (${}^4_2\text{He}$)	Fast electron (${}^0_{-1}\text{e}$)	Electromagnetic wave
Charge	+2 (Positive)	-1 (Negative)	No charge
Ionising Power	Maximum	Moderate	Minimum
Penetrating Power	Least (stopped by paper)	Moderate (stopped by aluminium)	Maximum (stopped by thick lead)
Deviation in E/M field	Deflected towards negative plate	Deflected towards positive plate	Not deflected

****3. Changes in the Nucleus Due to Radiation****

During alpha emission: Atomic number decreases by 2, mass number decreases by 4.

Example: ${}_{92}^{238}\text{U} \rightarrow {}_{90}^{234}\text{Th} + {}_2^4\text{He}$

During beta emission: Atomic number increases by 1, mass number remains unchanged. A neutron converts into a proton.

Example: ${}_{90}^{234}\text{Th} \rightarrow {}_{91}^{234}\text{Pa} + {}_{-1}^0\text{e}$

During gamma emission: No change in atomic number or mass number. The nucleus loses energy only.

****4. Uses and Hazards of Radioactivity****

****Uses:****

Medical: Sterilising equipment, radiotherapy for cancer, diagnosis (using radioisotopes like Cobalt 60, Iodine 131).

Industrial: Finding thickness of materials, detecting flaws in metal castings.

Power generation: Nuclear reactors.

****Harms:**** High radiation can kill cells, cause cancer, genetic mutations, and radiation burns.

****Safety Precautions:**** Protective lead aprons and walls, handling with tongs and wearing gloves, keeping source in lead lined containers, never looking directly at a radioactive source.

****5. Background Radiation****

The natural low level radiation present all around us from cosmic rays, radioactive rocks, soil etc.

****6. Nuclear Fission and Fusion****

****Nuclear Fission:**** The splitting of a heavy nucleus (like Uranium-235) into two lighter nuclei when bombarded with a slow neutron, releasing a tremendous amount of energy.

Example: ${}_{92}^{235}\text{U} + {}_0^1\text{n} \rightarrow {}_{56}^{141}\text{Ba} + {}_{36}^{92}\text{Kr} + 3{}_0^1\text{n} + \text{Energy}$

****Nuclear Fusion:**** The combining of two light nuclei to form a heavier nucleus at extremely high temperatures, releasing large energy.

Example: ${}_1^2\text{H} + {}_1^3\text{H} \rightarrow {}_2^4\text{He} + {}_0^1\text{n} + \text{Energy}$

****IMPORTANT QUESTIONS: RADIOACTIVITY****

****Q1.**** Define the following terms: Atomic Number, Mass Number, and Nuclide.

****Q2.**** What is radioactivity? Name the three types of radiations emitted by a radioactive element.

****Q3.**** Compare the ionising power and penetrating power of alpha, beta and gamma radiations.

****Q4.**** A radioactive nucleus emits a beta particle. What changes occur in the atomic number and mass number? Give an example with equation.

****Q5.**** Write the equation for the alpha decay of Uranium-238.

****Q6.**** State two uses and two hazards of radioactivity.

****Q7.**** Write three safety precautions to be taken while handling radioactive sources.

****Q8.**** Distinguish between nuclear fission and nuclear fusion. Give one example equation for each.

****Q9.**** What is background radiation? Name one source of it.

****Q10.**** Why are gamma rays used to treat cancer? On which property do we depend for this use?